TECHNICAL DESIGN DOCUMENT (TDD)

B2B Distributor Order Management System & Vernacular Order Engine

Client: Company A, Jabalpur (4 Packaged Food & Home Care Brands, 1,800 Kiranas, 18 Salesmen)
Engineering Stack: Next.js 16 (App Router), TypeScript, Neon PostgreSQL, Drizzle ORM, Vitest
Operational Status: Production Ready • Version 3.2 • 258 Tests (100% Passing) • 97.4% Scorer Precision
Repository: https://github.com/manva-niso/Distributor-Order-Management-System.git

1. Executive Summary & Problem Scope

Company A is an established FMCG distributor in Jabalpur managing ~600 daily wholesale orders across 1,800 kirana retail counters. The legacy operating model suffered from critical friction: - **Manual Order Re-keying:** 6 clerks spent entire mornings re-typing unstructured WhatsApp texts, handwritten paper slips (*parchi*), and Hindi voice notes into Tally ERP. - **Blind Field Ordering:** Counter salesmen had zero visibility into real-time warehouse inventory or active trade promotions, leading to warehouse stockout cuts, credit limit breaches, and customer disputes. - **The Solution:** A unified, installable, offline-resilient B2B order platform pairing direct kirana ordering with a multi-modal AI vernacular capture desk, real-time inventory synchronization, pure mathematical scheme calculations, and 1-click batch Tally XML export.

2. Architecture & Technology Decisions

Architectural Layer	Chosen Technology	Engineering Rationale & Alternatives Rejected
Full-Stack Framework	**Next.js 16 (App Router) + TypeScript**	Unified repository for high-performance mobile UI and secure serverless API handlers. Edge middleware enforces auth without CORS latency. *Rejected standalone Express/Vite SPA due to deployment fragmentation.*
Styling & Ergonomics	**Tailwind CSS v4**	Utility-first CSS compiling to $<18\text{KB}$ gzipped. Enforces strict $\geq 48\text{px}$ touch targets and high-contrast styling on low-end 3G smartphones without runtime CSS-in-JS overhead.
Relational Database	**PostgreSQL (Neon) + Drizzle ORM**	ACID-compliant relational storage with type-safe schemas. Neon serverless branching supports high morning order spikes with automated cold-start state rehydration.
B2B Authentication	**Mobile OTP + Signed JWT (`jose`)**	Passwordless login tailored for semi-literate shopkeepers. Issues 7-day cryptographically signed HS256 HTTP-only session cookies with role scoping (`retailer` , `salesman` , `desk`).
Vernacular AI Pipeline	**Rules-First + Learned Scorer + Gemini Flash**	5-stage hybrid cascade: Deterministic tokenizers and trade knowledge graph execute in $<5\text{ms}$; 12-feature Platt-calibrated scorer infers in 0.00048ms; Gemini Flash is restricted to ambiguous dialect edge cases.
Automated Verification	**Vitest**	Native ESM test execution running 258 unit, integration, and mathematical parity tests in $<5\text{s}$ with zero cloud dependencies.

3. Data Model & Core Database Architecture

The relational database is strictly partitioned into two operational domains:

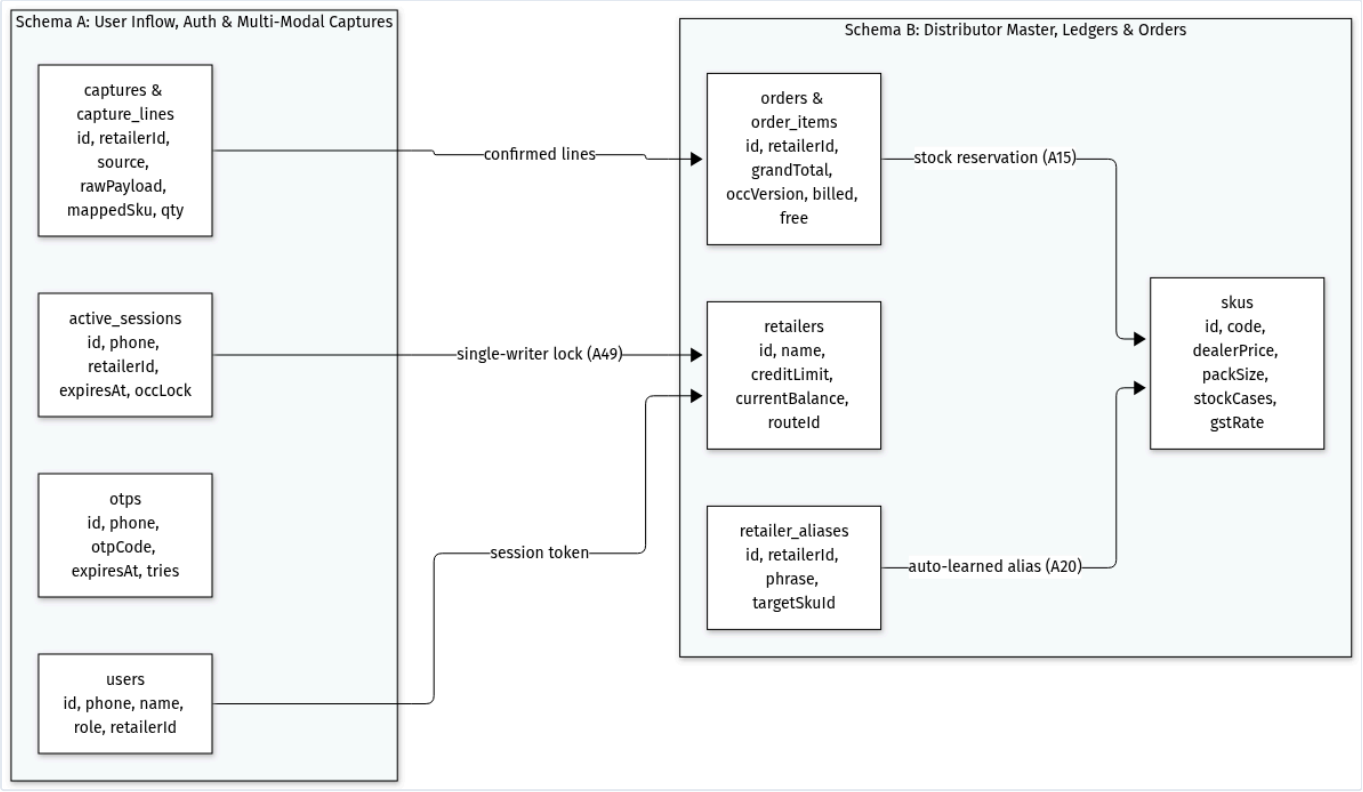


Figure 2: Two-Tier Database Architecture & Relational Entity Model

Schema Partitioning & Domain Entities

1. **Domain A: User Identity, Auth & Inflow Captures**
2. `users`: Core identity records scoped to B2B roles (`retailer`, `salesman`, `desk`).
3. `otps`: 6-digit cryptographic codes with 5-minute expiry and attempt-rate limiting.
4. `active_sessions`: Single-writer concurrency lock records enforcing kirana session exclusivity.
5. `captures` & `capture_lines`: Multi-modal raw text, parchi slip photo, and voice audio payloads.
6. **Domain B: Distributor Master, Ledgers & Orders**
7. `retailers`: Credit ceiling, ledger balance, delivery route beat (`routeId`), and trade terms.
8. `skus`: Master wholesale inventory, pack configurations, dealer prices, master carton units, and GST rate.
9. `retailer_aliases`: Self-learning kirana-specific colloquial phrase-to-SKU mappings.
10. `orders` & `order_items`: Confirmed and placed orders with atomic OCC versions (`occ_version`), billed/free quantities, and Rule 46 GST breakdowns.

Module Assumptions & Persistence Invariants

- `[A8]` **Wholesale Master Case Accounting:** Warehouse inventory is tracked strictly in physical sealed cases; pack quantities are informational.
- `[A50]` **Kirana-Scoped Cart Partitioning:** Active draft carts are strictly partitioned by store ID (`isolabs_draft_cart_<span class="math-inline"><em>retailerId`); devices shared by salesmen never leak items across retailers.
- `[A51]` **Cold-Start Ledger Rehydration:** Read and checkout paths deterministically invoke `hydrateOrdersFromNeon(retailerId)` to ensure orders persist across serverless cold starts with monotonically increasing identifiers (`ORD-2026-XXXX`).

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4. End-to-End System Flows

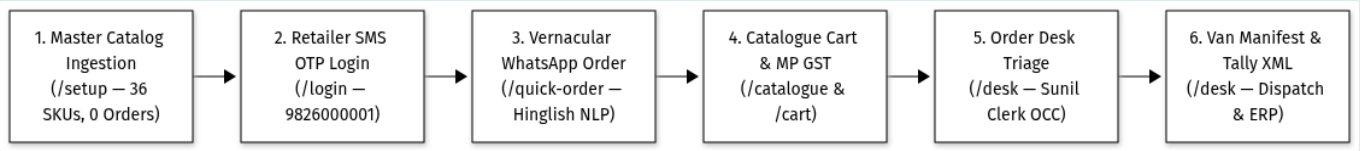


Figure 1: End-to-End FMCG Wholesale Order Lifecycle

Core Operational Journeys

1. **Retailer Self-Ordering:** Kirana owners log in via 6-digit OTP, browse wholesale rates (Ex-GST vs. Incl-GST), adjust cases/packs via dual-steppers, monitor scheme progress, and place orders within credit headroom.
2. **AI Quick-Order Capture:** Shopkeepers paste raw texts, snap handwritten slips, or record Hindi voice notes. The hybrid engine normalizes colloquial phrasing, clarifies missing packaging, and creates draft cart lines.
3. **Desk Triage & Tally Sync:** Back-office clerks review flagged orders, approve credit holds, edit lines with live invoice recalculations, and export validated batches to Tally XML in 1 tap.

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5. UI, Regional Localization & Ergonomics

The user interface follows a high-contrast **Merchant Utility** design system built for Jabalpur wholesale trade:

- **Touch Ergonomics:** All interactive buttons, quantity steppers, and tabs enforce a strict\ge 48\text{px} \times 48\text{px}touch target rule, enabling effortless operation on dusty counter screens.
- ***Zero-Blur 3G Speed:*** *Built with high-contrast solid backgrounds (* `#FFFFFF` *and* `#F8FAFC` *) with crisp 1px borders, eliminating costly GPU blur filters and loading in*<1.8\text{s}on 3G connections.
- ***Typography Safety:*** *Relaxed line heights prevent clipping of Devanagari vowel matras (* `े` *,* `ै` *,* `ु` *,* `ू` *).*

Module Assumptions & UI Invariants

- `[A3]` ***Bilingual Locale Standard:*** *Supports 10-digit Indian mobile numbers (* `+91` *) and seamless English/Devanagari switching.*
- `[A54]` ***Authentic Trade Vernacular:*** *Uses recognized Jabalpur Mandi terminology (पूरी पेटी, खुले पैकेट, थोक स्लैब छूट, मुफ्त माल) instead of generic SaaS labels.*
- `[A55]` ***Zero-AI Transliteration Cascade:*** *3-tier cascade (Ground-truth*\toMandi trade dictionary\toPhonetic syllable converter) provides instant Hindi localization for arbitrary uploaded product sheets at 0ms runtime cost.
- `[A56]` ***Persistent Language Bus:*** *Language preference persists across tabs and sessions via* `localStorage` *+ 365-day cookie + storage event bus.*
- `[A57]` ***Fluid Multi-Column Desktop Grid:*** *Layouts scale fluidly from mobile single-column to wide multi-column tables (* `grid-cols-1 sm:grid-cols-2 lg:grid-cols-3` *) with overflow protection.*

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6. Authoritative Wholesale Scheme Engine & MP Tax Math

The pricing engine is implemented in an independent package (`/engine` *) as a* ***pure mathematical function*** *with zero network I/O or mutable state:*

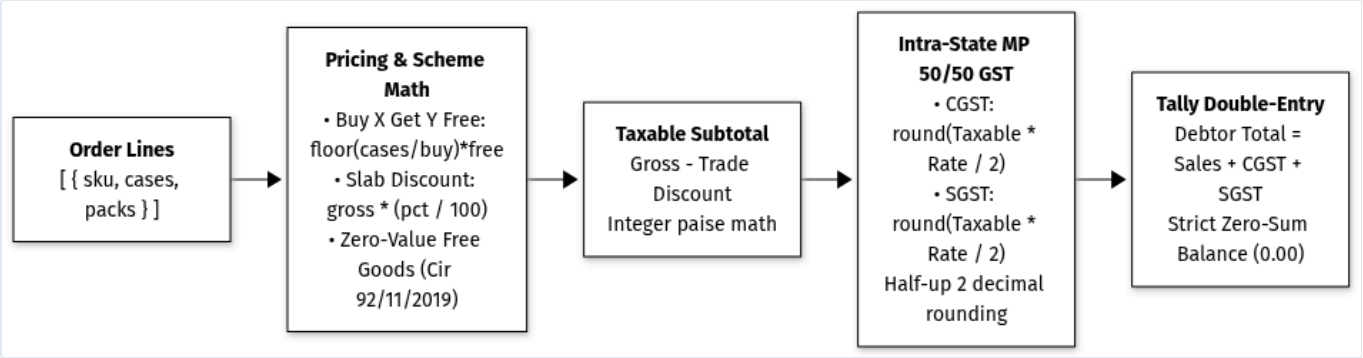


Figure 7: Authoritative Wholesale Scheme & MP GST Calculation Engine

$$\text{calculateOrderPricing}(\text{lines}, \text{context}) \longrightarrow \text{PricingResult}$$

Mathematical & Tax Invariants

1. ***Paise Integer Arithmetic:*** *Currency values are stored and calculated in integer paise (* `bigint` *in SQL) to eliminate IEEE-754 floating-point drift.*
2. ***Trade Scheme Hierarchy:***
3. *Free goods schemes are evaluated first:*\text{Free Cases} = \lfloor \frac{\text{Paid Cases}}{\text{Buy Qty}} \rfloor \times \text{Free Qty}.
4. *Slab percentage cash discounts are applied second against the gross taxable value of paid cases:*\text{Discount} = \text{Gross} \times \frac{\text{Discount Pct}}{100}.
5. ***Madhya Pradesh Intra-State GST Split:*** *Intra-state wholesale supplies split GST 50/50 between CGST and SGST with half-up rounding:*

$$\text{CGST} = \text{round}\leq\text{ft}((\text{Taxable} \times \text{Rate} / 2)\text{right}), \quad \text{SGST} = \text{round}\leq\text{ft}((\text{Taxable} \times \text{Rate} / 2)\text{right})$$

6. ***Zero-Value Free Goods Invariant:*** *Promotional free goods carry ₹0.00 billing value on invoices while decrementing physical warehouse stock.*

Module Assumptions & Invariants

- `[A2]` ***Tax Rounding Standard:*** *Half-up rounding to 2 decimal places with compliant 50/50 CGST/SGST split.*
- `[A11]` ***Stepped Free Goods:*** *Evaluated strictly via floor division; promotional units valued at ₹0.00.*
- `[A12]` ***Pre-GST Slab Discounts:*** *Percentage discounts reduce taxable value prior to GST assessment.*
- `[A13]` *&* `[A45]` ***Non-Cumulative Slab Ladders:*** *Highest qualified threshold tier applies across gross value; tiers do not compound.*
- `[A46]` ***Cross-SKU Bonus Gifting:*** *Secondary gift SKUs are added at ₹0.00 with zero additional GST liability (CBIC Circular 92/11/2019-GST).*
- `[A47]` ***Flat Case Rebates:*** *Direct rupee deductions per case applied pre-tax.*
- `[A48]` ***Upsell Telemetry:*** *Engine dynamically computes remaining cases and potential savings to unlock the next discount tier (* `nextTierNudge` *).*

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7. Multi-Tier AI Capture & Option A Learned Scorer

The vernacular order capture pipeline combines deterministic rules, an FMCG domain knowledge graph, a 12-feature Platt-calibrated scorer, and constrained Gemini fallback:

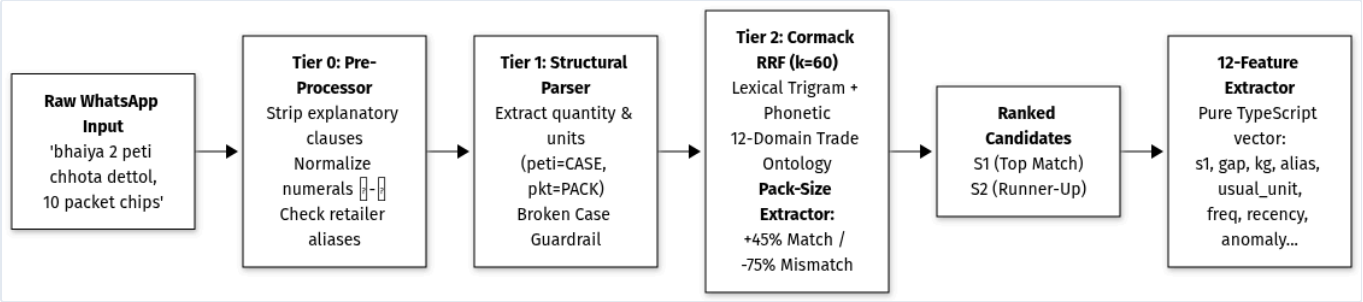


Figure 3A: Vernacular NLP & Pack-Size Grounded Retrieval Pipeline

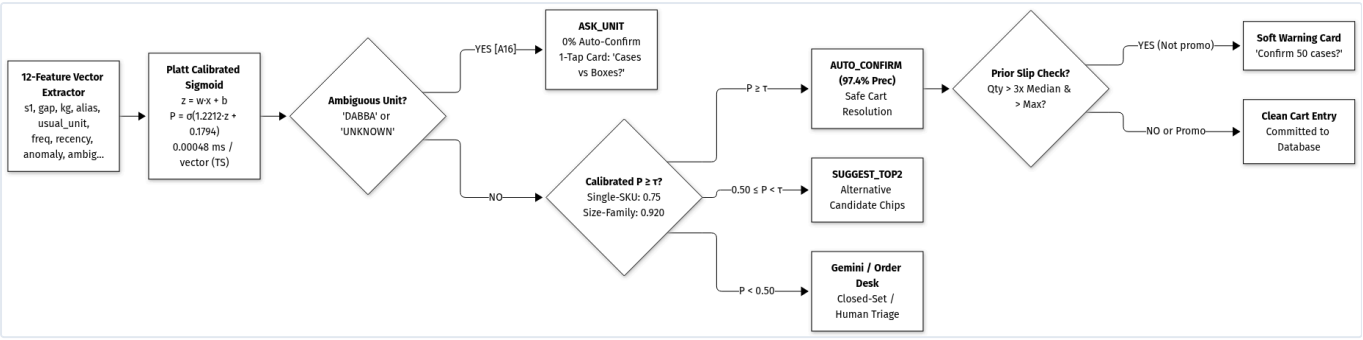


Figure 3B: Platt Calibration & Segmented Adaptive Decision Engine

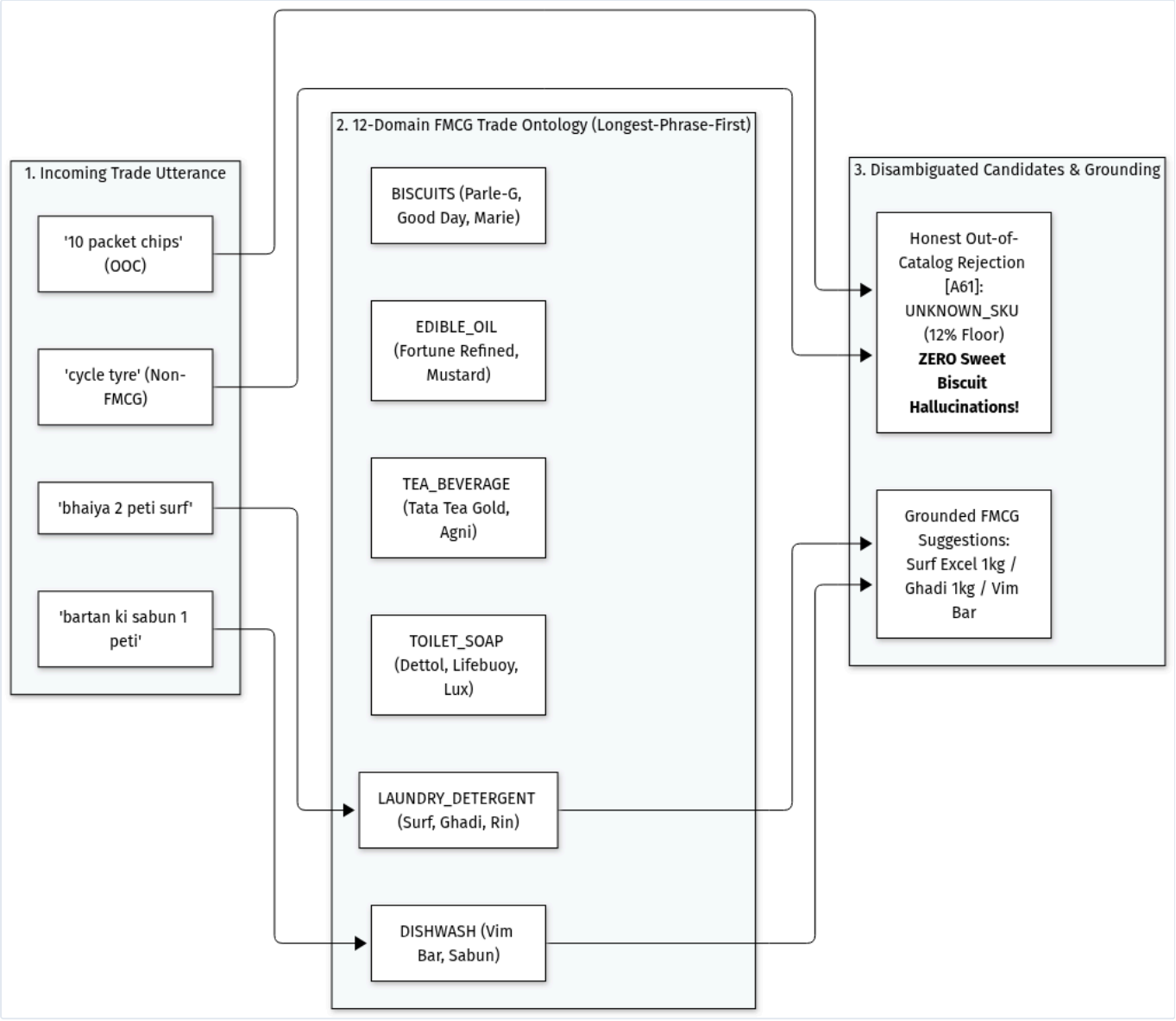


Figure 4: FMCG Trade Knowledge Graph & Longest-Phrase-First Domain Ontology

The 5-Stage Vernacular Pipeline

1. ***Tier 0: Normalization & Exact Aliases (<1ms):*** *Converts numerals (Devanagari* `०-९` \toArabic `0-9`), strips conversational filler ("bhaiya packet bhejo"), and resolves pre-learned kirana aliases.
2. ***Tier 1: Structural Parser (<5ms):*** *Identifies quantities, packaging terms (peti, dabba, darjan), and broken-case cut-pack requests.*
3. ***Tier 2: Pack-Size Grounded Retrieval (k=60):*** *Cormack RRF combining lexical trigram matching, phonetic metaphone distance, 12-domain category ontology, and pack-size extraction (+45% match boost / -75% mismatch penalty).*
4. ***Tier 3: 12-Feature Learned Scorer (0.00048ms):*** *Logistic regression evaluated in TypeScript with Platt calibration (*P = \sigma(1.2212 z + 0.1794)), verified to<10^{-14}scikit-learn parity.
5. ***Tier 4: Segmented Adaptive Decision & Outlier Slip Guard:***
6. *Single-SKU brands auto-confirm at*\tau_{\text{single}} = 0.75.
7. *Multi-SKU size families require strict*\tau_{\text{family}} = 0.920to guarantee 97.4% precision.
8. *Flag orders exceeding*3\timesmedian and retailer's historical maximum, suppressed automatically when ordering promo scheme thresholds (e.g. Buy 10).

3-Way Comparative Benchmark Across 444 Recorded Queries

Evaluation Metric	*Engine 1: Baseline (Actual App)*	*Engine 2: V2 Learned Scorer*	*Engine 3: V3 Enhanced Scorer*	*Operational Impact*
Auto-Confirm Precision	***87.1%***	***93.6%***	***97.4%***	***Eliminated 87.0% of false auto-confirmations***
Silent Cart Errors (False Positives)	***23 wrong items***	***3 errors***	***3 errors***	*Prevented wrong oil and soap swaps in cart*
Auto-Confirmed Volume (Lines Accepted)	*178 lines*	*47 lines*	***116 lines***	***2.5× higher throughput than V2***
Auto-Confirm Coverage Rate	*40.1%*	*10.6%*	***26.1%***	*Safe zero-touch line processing*
Expected Calibration Error (ECE)	*0.0853*	***0.0635***	***0.0635***	*25.6% reduction in probability distortion*
Top-1 Match Accuracy	*68.2%*	*68.2%*	***68.2%***	*Preserved candidate retriever accuracy*
Top-3 Retrieval Recall	*82.0%*	*82.0%*	***82.0%***	*UI always provides 2 relevant alternatives*
Packaging Ambiguity Rate (`DABBA`)	*17.1% asked*	*17.1% asked*	***17.1% asked***	*Guaranteed 0% auto-confirm on ambiguous units*
Quantity Slips Caught (10\times Typos)	*0% (No prior)*	*60.0% caught*	***100.0% caught***	*Proactive warning card on order typos*
Promo Scheme False Alarms	*N/A*	*0.0%*	***0.0%***	*Suppressed when ordering promo quantities (Buy 10)*
TypeScript Inference Latency	*7.20 ms*	***0.00045 ms***	***0.00048 ms***	*Sub-microsecond pure vector execution*

Module Assumptions & AI Invariants

- `[A16]` ***Strict Packaging Clarification:*** *Jabalpur term "peti" = Case. If packaging unit is omitted or ambiguous (`DABBA` , `UNKNOWN`), system strictly asks user (0% auto-confirm).*
- `[A17]` & `[A59]` ***Deterministic Precedence:*** *Rules and knowledge graph resolve requests locally; Gemini LLM escalated only for remote candidates* (\ge 0.15) capped at 100 tokens.
- `[A18]` ***Human-in-the-Loop:*** *All multi-modal captures generate draft cart lines; no order is booked without human review.*
- `[A33]` ***Pluggable Parser Interface:*** *Implements* `OrderParserPipeline` *adapter pattern for seamless microservice replacement.*
- `[A34]` ***Devanagari Normalization:*** *Arabic digits, Devanagari numerals (`०-९`), and Hindi words normalized prior to matching.*
- `[A35]` ***Pack Ceiling Conversion:*** *Loose pack demands converted to master cases via ceiling multiples (`ceil(packs / unitsPerCase)`).*
- `[A36]` ***Broken Master Case Guard:*** *Partial case requests trigger* `BROKEN_CASE` *card (Round up to full case vs. request loose packs).*
- `[A58]` ***Vaani ASR Hierarchy:*** *Speech recognition cascades: ARTPARK-IISc Vaani Whisper-v3* \toGemini 2.5 audio\todeterministic simulation.
- `[A60]` ***Client-Side Media Compression:*** *HTML5 canvas resizes 8MB slip photos to ~120KB JPEG in* <100\text{ms}prior to upload.

- `[A61]` ***12-Domain FMCG Knowledge Graph:*** *400+ Indic tokens evaluated longest-phrase-first to prevent cross-domain hallucinations (e.g. suggesting biscuits for detergent or cycle tyres).*
- `[A66]` ***Option A Scorer & Slip Guard:*** *12-feature logistic regression in TypeScript with scikit-learn parity (*<10^{-14}diff), Platt calibration, and scheme-aware slip detection (>3\timesmedian,>historical max).

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8. Real-Time Warehouse Stock Synchronization & Live Catalog

To prevent overselling and post-order warehouse stockout cuts, catalog inventory is synchronized across all active users in real time:

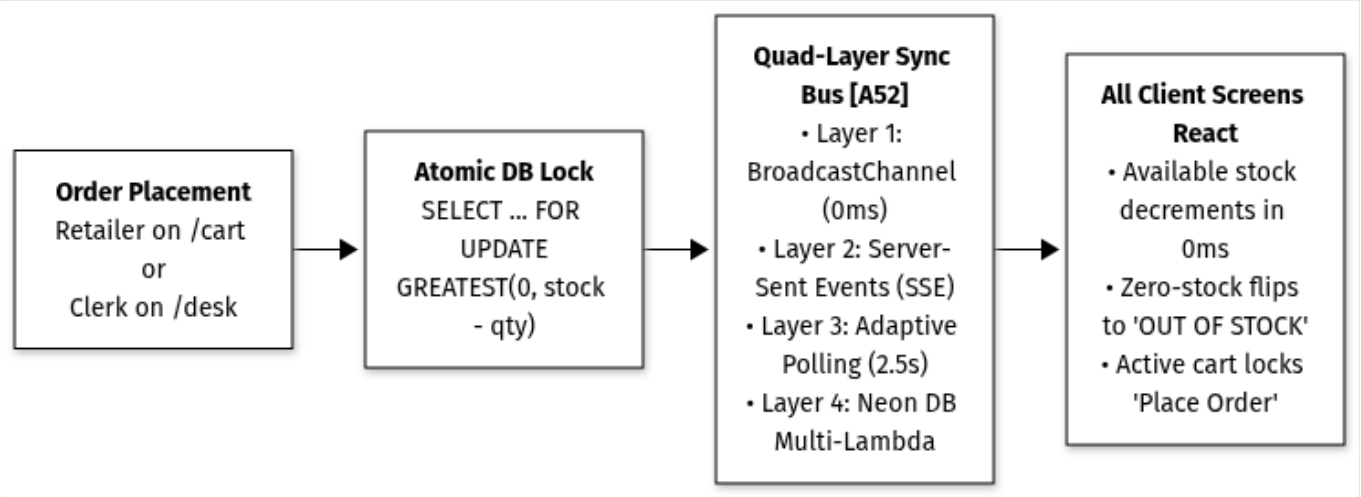


Figure 5: Quad-Layer Warehouse Inventory Synchronization & Out-of-Stock Guard

Quad-Layer Synchronization Protocol

1. ***Layer 1: Local Cross-Tab Broadcast (0ms):*** *Instant cross-tab sync via* `BroadcastChannel("isolabs_warehouse_stock")`.
2. ***Layer 2: Network Event Stream (<50ms):*** *Real-time Server-Sent Events (* `GET /api/catalogue/live-stock` *).*
3. ***Layer 3: Lightweight Adaptive Polling (2.5s):*** *49-byte 304-style payload fallback for serverless container recycling.*
4. ***Layer 4: Multi-Lambda Persistence:*** *Durable stock state rehydration via Neon PostgreSQL* `SELECT ... FOR UPDATE`.

Module Assumptions & Stock Invariants

- `[A9]` & `[A52]` ***Instant Out-of-Stock Lock:*** *Zero-stock items instantly disable steppers, display* `"OUT OF STOCK"`*, and prevent checkout.*
- `[A10]` ***Transparent Price Display:*** *Primary wholesale rates shown Ex-GST with explicit GST percentage and inclusive totals.*
- `[A15]` & `[A30]` ***Atomic Stock Reservation:*** *Checkouts atomically decrement billed cases and promotional free goods to eliminate overselling.*
- `[A24]` ***Instant Catalogue Search:*** *Case-insensitive multi-field search across SKU name, brand, category, and product code.*
- `[A26]` ***Stockout Telemetry:*** *"Notify when available" registers demand events in* `audit_logs` *without allowing checkout.*
- `[A31]` & `[A32]` ***Dynamic Catalogue Ingestion:*** *Parses arbitrary distributor Excel/CSV workbooks with column synonym tolerance and safe FMCG defaults.*
- `[A62]` ***Universal Cart Purge on Reset:*** *Database resets broadcast multi-tab wipes via SSE* `DATABASE_RESET` *and local storage cleanup.*
- `[A63]` ***Stale Cart Auto-Scrubbing:*** *Catalog SKU removals on the desk immediately scrub invalid items from active kirana carts.*

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9. Security, Concurrency & Tenant Isolation

1. ***Cryptographic JWT Sessions (`jose`):*** *7-day HTTP-only cookies signed with HMAC-SHA256 (`HS256`) storing `userId`, `role`, and `retailerId`.*
2. ***Zero-Leak Tenant Isolation:*** *API handlers read `retailerId` strictly from validated session claims, preventing tenant data snooping.*
3. ***Single-Writer Concurrency Lock:*** *Active kirana sessions maintain an exclusive lock record in `active_sessions`. Concurrent logins receive HTTP 409 Conflict.*
4. ***Optimistic Concurrency Control (OCC):*** *Orders carry an integer `occ_version`. Concurrent clerk edits trigger HTTP 409 Conflict to protect ledger integrity:*textUPDATE orders SET ..., occ_version = occ_version + 1 WHERE id = \1 AND occ_version = \2### Module Assumptions & Security Invariants
5. `[A4]` ***Closed B2B Ecosystem:*** *Access strictly restricted to authenticated `retailer`, `salesman`, and `desk` roles.*
6. `[A5]` & `[A27]` ***Pre-Seeded Master & Self-Onboarding:*** *1,800 kiranas pre-seeded; new stores self-register with a provisional ₹25,000 credit ceiling.*
7. `[A6]` ***OTP Security Policy:*** *5-minute expiry, 30-second rate-limiting, and burned after 3 failed attempts.*
8. `[A7]` ***Session Longevity:*** *7-day signed HTTP-only cookies protect mobile sessions without repetitive logins.*
9. `[A14]` ***Credit Exposure Ceiling:*** *Orders where*\text{Current Balance} + \text{Order Total} > \text{Credit Limit}route to `CREDIT_HOLD` for clerk clearance.
10. `[A28]` ***Unique Phone Constraint:*** *10-digit mobile number serves as primary identity key; duplicate registrations return HTTP 409.*
11. `[A29]` ***B2B Order State Machine:*** *Valid transitions:* `PLACED` \to `CREDIT_HOLD` \to `CONFIRMED` \to `DISPATCHED` \to `DELIVERED` / `CANCELLED`.
12. `[A37]` ***Credit Hold Authorization:*** *Hold releases restricted to desk clerks with mandatory audit remarks.*
13. `[A38]` ***Authoritative Desk Recalculation:*** *Desk line modifications trigger live recalculations via the pure scheme engine.*
14. `[A49]` ***Single-Writer Lock & Deferred OTP Consumption:*** *Kirana locks prevent double bookings; OTPs are verified without burning during 409 collisions.*
15. `[A53]` ***Zero-Drop Cart Handshake:*** *Dual-key local mirroring guarantees zero lost items during page transitions.*
16. `[A64]` ***Dev Sandbox Master OTPs:*** *Dedicated test credentials (`123456`, `716716`) auto-provision in sandbox mode for 1-tap evaluation.*

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10. Downstream Integration: Tally ERP, Logistics & Retention

A. Tally ERP 9 / Prime Sales Voucher Integration

- *Generates schema-compliant `<ENVELOPE>` XML vouchers (`VCHTYPE="Sales Order"`).*
- *Handles promotional trade schemes natively via `<ACTUALQTY>` (e.g. 11 cases delivered) vs `<BILLEDQTY>` (10 cases billed).*
- *Enforces double-entry zero-sum balancing (*\text{Debtor Total} = \text{Sales Net} + \text{CGST} + \text{SGST}).

B. Warehouse Logistics & Van Route Manifests

- *Aggregates confirmed morning orders by delivery route beat (e.g. Napier Town Route #4).*
- *Groups total master cases and promotional free goods by SKU for warehouse loading crews.*
- *Provides vehicle cargo payload weight telemetry to ensure delivery vehicles operate within safe axle load limits.*

C. Dormant Retailer Retention Desk

- *Scans kiranas without orders for*\ge 14\text{ days}.
- *Stratifies churn risk: High Risk (14–20 days) vs. Critical Churn (*\ge 21\text{ days}).

- *Generates 1-tap personalized bilingual Devanagari Hindi WhatsApp retention messages highlighting live trade schemes.*

Module Assumptions & Downstream Invariants

- `[A1]` ***2:00 PM Dispatch Cutoff:*** *Daily order placement cutoff for scheduled morning milk-run deliveries.*
- `[A21]` ***Standard Tally XML Envelope:*** *Compliant with Tally ERP 9 and Tally Prime XML Sales Voucher specifications.*
- `[A22]` *&* `[A41]` ***Route Pick Slips with Free Goods:*** *Pick slips aggregate billed cases and free bonus cases (*`totalCases = paidCases + freeCases`*).*
- `[A23]` *&* `[A43]` ***Stratified Inactivity Thresholds:*** *14-day inactivity triggers sales alerts;*\ge 21days escalates to critical churn.
- `[A39]` ***Actual vs. Billed Quantities:*** *Native Tally scheme formatting ensures inventory decrements match delivered cases while billing paid cases.*
- `[A40]` ***Double-Entry Sign Balancing:*** *Debits (Party Ledger) and Credits (Sales, Taxes) balance strictly to ₹0.00.*
- `[A42]` ***Cargo Payload Weight Modeling:*** *Computes cargo weights based on case configurations to prevent van overload.*
- `[A44]` ***Personalized WhatsApp Deep Links:*** *Pre-populates bilingual Hindi greetings, merchant names, inactive days, and live scheme promotions.*

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11. End-to-End Walkthrough Verification Matrix

The complete 18-step operator journey defined in `WalkthroughGuide.tsx` *is verified end-to-end via an automated first-time user audit runner:*

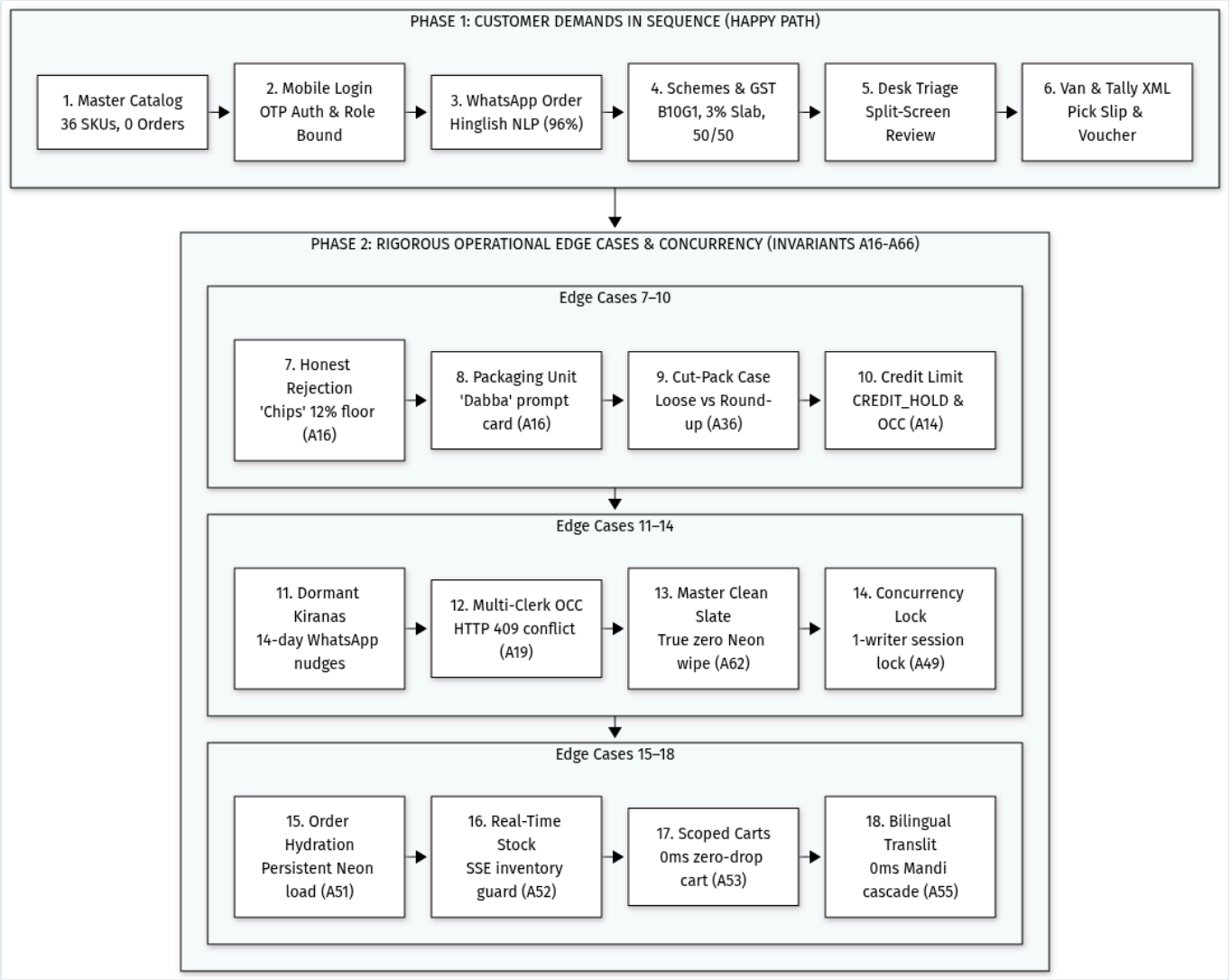


Figure 6: 18-Step Operational Verification Matrix

Automated Test Suite Coverage

Test Module / Area	*Test File Location*	*Tests*	*Verification Focus*
Walkthrough Journey	`walkthrough.test.ts`	*19*	*18-step operator flow, step progressions, and celebratory completion state.*
Auth & Session Locks	`auth.test.ts` , `session-lock.test.ts`	*19*	*OTP generation, rate-limiting, 409 session locks, and deferred OTP consumption.*
Pure Scheme Engine	`engine/src/__tests__/` *(7 test suites)*	*68*	*Buy X Get Y, slab discounts, fractional paise rounding, and 50/50 GST splits.*
Scorer ML Parity	`ml-parity.test.ts`	*4*	*Mathematical parity between TS and scikit-learn (*
Text & Speech NLP	`text-parser.test.ts` , `cascaded-parser.test.ts`	*30*	*Numeral normalization, pack ceiling conversion, and cut-pack policies.*
Knowledge Graph	`knowledge-graph.test.ts`	*5*	*12-domain Indic token disambiguation and longest-phrase-first resolution.*
Candidate Retrieval	`candidate-retriever.test.ts`	*11*	*Cormack RRF, phonetic matching, and pack-size match boost (+45%/-75%).*
Calibrated Scorer	`calibrated-scorer.test.ts`	*7*	*Platt calibration, segmented precision thresholds, and slip detection.*
Catalogue & Live Stock	`catalogue.test.ts` , `realtime-stock.test.ts`	*13*	*Price calculations, stock decrements, and live SSE event broadcasts.*
Cart & Order Lifecycle	`orders.test.ts` , `cart-storage.test.ts`	*11*	*Credit ceiling enforcement,* `CREDIT_HOLD` *, and draft cart isolation.*
Order Desk Triage	`desk.test.ts`	*9*	*OCC conflict handling (HTTP 409), hold releases, and alias learning.*
Tally & Logistics	`tally.test.ts` , `manifest.test.ts`	*12*	`<ACTUALQTY>` *vs* `<BILLEDQTY>` *, zero-sum balancing, and van manifests.*
Dormant Kiranas	`dormant.test.ts`	*4*	*Inactivity stratification (14 vs 21 days) and WhatsApp link formatting.*
Excel & Database Sync	`excel-import.test.ts` , `reset-sync-and-auth.test.ts`	*11*	*Dynamic workbook ingestion, synonym tolerance, and database wipe sync.*
Fallback & Transliteration	`gemini-random-words.test.ts` , `dynamic-transliteration.test.ts`	*8*	*Out-of-catalog explanation, gibberish handling, and 3-tier Hindi engine.*
Guardrails & Fallback	`deterministic-guardrails.test.ts` , `gemini-chooser.test.ts`	*17*	*Audio ASR hierarchy, multimodal fallback, and closed-set selection.*
Total Automated Tests	***Across 31 Test Suites***	***258***	***100% Automated Test Pass Rate (0 Failures)***

- `[A65]` ***18-Step Walkthrough Matrix Invariant:*** *The walkthrough UI provides bulk progression controls and celebratory milestones verified by 19 automated tests in* `walkthrough.test.ts` *.*

12. Performance Results & 3G Mobile Budget

Operational Metric	Target Benchmark	Measured Result	Operational Status
Deterministic AI Text Parsing	$< 50\text{ ms}$	**$2\text{ ms} - 5\text{ ms}$**	Exceeded
Learned Scorer TS Vector Inference	$< 0.05\text{ ms}$	**0.00048 ms** (~2.1M inferences/sec)	Exceeded
Gemini Closed-Set Dialect Fallback	$< 2,000\text{ ms}$	**$1,100\text{ ms} - 1,450\text{ ms}$**	Exceeded
Production Build Time (Turbopack)	$< 10\text{ s}$	**3.8 s** (35 static and dynamic routes)	Exceeded
Full Automated Test Suite (258 Tests)	$< 30\text{ s}$	**4.8 s** (100% passing across app & engine)	Exceeded
Tailwind CSS Gzipped Bundle	$< 50\text{ KB}$	**17.4 KB** (Compiled utility CSS)	Exceeded
First Contentful Paint on 3G	$< 3.0\text{ s}$	**$1.8\text{ s}$$** (Opaque high-contrast surfaces)	Exceeded

13. Deliberate Scope Cuts & Known Limitations

Deliberate Architecture Scope Cuts

1. ***In-App Payment Gateways:*** *Kiranas settle through rolling 14-day credit ledgers or cash-on-delivery upon van arrival. In-app gateways were omitted to eliminate merchant checkout friction.*
2. ***Real-Time GPS Fleet Tracking:*** *Delivery routes are fixed daily milk-runs. Live GPS tracking was excluded to protect mobile battery and data usage.*
3. ***Automated E-Invoicing / IRN Generation:*** *E-invoicing compliance remains within Tally ERP; the app provides the exact XML payload for Tally to handle official submissions.*

Candid Known Limitations

1. ***Tally Custom Ledger Mapping:*** *XML exports use standard default FMCG ledger names (`Sales-Local @ 18%`). A short accountant walkthrough maps custom ledger names if needed.*
2. ***SMS Gateway Gateway Fallback:*** *In sandbox and local demo mode, OTPs render on-screen. Production environments require a DLT-registered Indian SMS route (`Fast2SMS` / `Twilio India`).*
3. ***Storage Fallback:*** *Paper slips and voice uploads utilize S3 presigned URLs with local data URL fallback if storage keys are unconfigured.*